

Diffie-Hellman Key Exchange

Aim

To implement the Diffie-Hellman key exchange, which lets two parties agree on a shared secret over an insecure channel without ever transmitting it. Both agree on a public prime modulus p and generator g ; each keeps a private exponent and exchanges only a public value. Raising the other party's public value to one's own private exponent yields the same shared secret on both sides, because $(g^a)^b = (g^b)^a \bmod p$. Security rests on the difficulty of the discrete logarithm problem.

Algorithm

Diffie-Hellman is a key agreement, not an encrypt/decrypt cipher, so its two core methods are computing a party's public value and deriving the shared secret from the other party's public value.

```
1 Function PublicValue(g, secret, p):  
2   return (g raised to secret) mod p           // fast modular exponentiation
```

```
1 Function SharedSecret(other_public, my_secret, p):  
2   return (other_public raised to my_secret) mod p
```

The full exchange between Alice (private a) and Bob (private b):

```
1 Alice and Bob agree on public p (prime) and g (generator)  
2  $A \leftarrow \text{PublicValue}(g, a, p)$            // Alice sends A to Bob  
3  $B \leftarrow \text{PublicValue}(g, b, p)$            // Bob sends B to Alice  
4  $s_A \leftarrow \text{SharedSecret}(B, a, p)$        // Alice computes B raised to a  
5  $s_B \leftarrow \text{SharedSecret}(A, b, p)$        // Bob computes A raised to b  
6  $s_A$  equals  $s_B$  = the shared secret
```

Output

Exchange — public parameters $p = 23$, $g = 5$; Alice's private $a = 6$, Bob's private $b = 15$

Each side computes its public value and sends only that number across the wire:

Party	Private	Public value = (g raised to private) mod p	Sends
Alice	$a = 6$	$(5 \text{ raised to } 6) \bmod 23 = 8$	$A = 8$
Bob	$b = 15$	$(5 \text{ raised to } 15) \bmod 23 = 19$	$B = 19$

Each side then raises the received value to its own private exponent:

Party	Computes	(received raised to private) mod p	Result
Alice	$B^a \bmod p$	$(19 \text{ raised to } 6) \bmod 23$	2
Bob	$A^b \bmod p$	$(8 \text{ raised to } 15) \bmod 23$	2

Both arrive at the same value, so no decryption step is needed; the agreement itself is the output.

Result: shared secret established = 2

The private exponents a and b are never sent; an eavesdropper sees only p , g , $A = 8$ and $B = 19$, and recovering the secret would require solving the discrete logarithm.